

Exploring the Trade-offs between Energy and Performance of Federated Learning Algorithms

– PhD Progress Report 3rd year –

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What is Energy Consumption in Federated Learning?

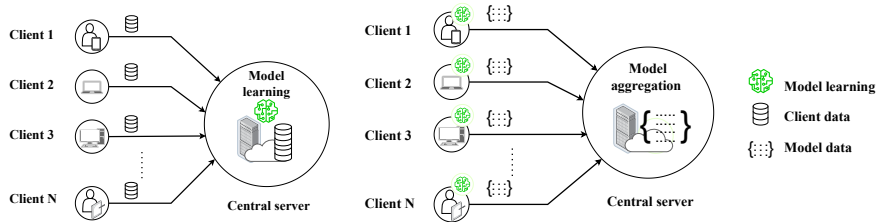


Figure 1: Left: Traditional Machine Learning: Data Comes to the Model → Data privacy issues.
Right: Federated Learning framework: Model Goes to the Data → Keep data privacy.

Energy consumption

- Computation: distributes computation across many clients
- Communication: frequent lightweight model updates.

Thesis Context

- This thesis is one part of **ANR DELIGHT** (aDvancing fEderated LearnInG while reducinG tHe carbon fooTprint) project.
- Target of **DELIGHT**: improve FL performance but also incorporate **energy efficiency** as **one of the metrics of FL** to push FL towards sustainability.
- Target of **my thesis**: develop an automatic framework to explore the trade-offs between Energy and FL performance.

Thesis initial plan

Phases of the thesis:

- Set up an experimental environment on Grid'5000¹ (g5k) to gather performance and energy metrics.
 - ▷ Create a use-case for the Flower framework.
 - ▷ Build a reproducible and automated framework for obtaining metrics for this use case.
- Propose, formulate energy model, and implement the different leverages.
- Explore the impact of the leverages on both energy and performance.

¹a distributed testbed system (multiple clusters, nodes, cores,...).

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1st year

From April 2024 to the end of August 2024:

- Conducted a survey of related work from 2019 to 2024 (31 papers).
- Carried out initial experiments on energy measurement in a distributed system (Grid'5000).

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2nd year

From August 2024 to August 2025:

- Prepared a survey paper and revised it for submission (included 50 articles - selected from 263 screened articles, focus on energy and experimental design for FL)
- Developed FedE-ator (version 0), a framework for measuring energy consumption of FL in distributed systems.

Framework functions	Framework limitations
• DVFS-based CPU or GPU frequency control • Multi-instance execution and repetitions • Flexible FL configuration	• No CPU+GPU heterogeneous support • Only validate on Flower²

▶ Go

²A FL framework: <https://flower.ai/docs/framework/index.html>

Plan for 3rd year

Target / To-do list	Current status
Framework heterogeneity: Develop a heterogeneous framework supporting both CPU and GPU nodes within the same experiment.	Completed
Framework reproducibility: Package the environment with Nix, allowing users to reproduce experiments and results reliably.	In progress (50%), Documentation completed; Nix packaging pending.
Scientific evaluation: Extend experiments with various FL frameworks and settings to investigate the trade-off between accuracy and energy consumption.	Completed
Resource optimization: Investigate real-world optimization and resource allocation by integrating optimization algorithms into the FL system together with energy measurement.	In progress (50%)
Publications: Submit and publish research papers.	In progress (50%), 1 workshop paper accepted, 1 journal paper under second-round review.

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Survey

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Survey paper

- Title: **Towards Sustainable Federated Learning: A Survey on Energy and Experimental Practices**³
- Submitted to: Sustainable Computing: Informatics and Systems Journal⁴
 - ▷ First submission on 29/08/2025
 - ▷ Reviewer's feedback *main revision*, on 14/05/2026 (mon anniversaire)
 - ▷ Revised and re-submission on 11/06/2026

³See more [▶ Research Questions](#)

⁴<https://www.sciencedirect.com/journal/sustainable-computing-informatics-and-systems>

FedE-ator: Overview & Architecture

Three-layer architecture:

- 1 **Layer 1 - FedE-ator Core:** handles input/output, orchestrates experiment campaigns
- 2 **Layer 2 - Expetator^a (Monitoring):** uses both SW and HW methods, synchronized via shared timestamps
- 3 **Layer 3 - FL Module^b:** SSH-based execution on server/client nodes

► Limitations ► Features

^aCustomized branch extending Expetator, supporting adaptive CPU/GPU monitoring for heterogeneous environments.

^bExtended version supporting different FL frameworks

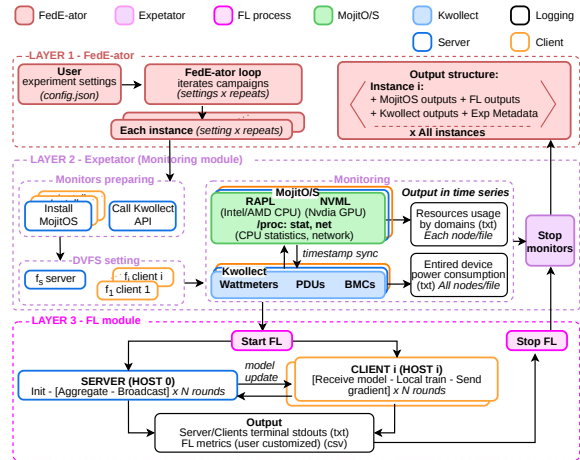


Figure 2: FedE-ator architecture

Experimental Setup

Platform: Grid'5000 (nova (CPU), gemini (GPU), gros (CPU) clusters)

Benchmark:

- **FL frameworks:** Flower & FedScale
- **Model:** CustomCNN (LeNet5), MobileNetV2, ResNet18
- **Dataset:** CIFAR-10 (IID, fixed seed)
- **FL strategy:** FedAvg (all clients participate each round)

Scenarios:

- **SC1 - Heterogeneous:** CPU+GPU nodes, 3 [CPU, GPU] freqs, 2 repetitions.
- **SC2 - Large-scale:** 90 CPU nodes⁵, $f_{\min}$ & $f_{\max}$, stability test.
- **SC3 - Cross-framework:** Flower vs. FedScale, 3 models, 4 freqs, 5 repetitions.

⁵In the 2nd year, I only tested with 20 nodes for a large-scale scenario

SC1 - Heterogeneous

Table 1: Settings

Server: nova-10
Clients: gemini-[1,2], nova-[11,12]
Freq. GHz: $f_{\min}$: [0.135, 1.2], $f_{\max}$: [1.53, 2.1]
Framework: Flower
Strategy: FedAvg
Library: PyTorch
Model: CustomCNN
Local Epochs: 10
Rounds: 3
Repetitions: 2

Table 2: Results

Freq. [GPU, CPU] (GHz)	[0.135, 1.2]	[0.832, 1.7]	[1.53, 2.1]
Time monitored (s)	250	158	134
Time FL training (s)	231	145	112
Mean P-entire system (W)	350	359	377
Mean P [GPU] (W)	752.22	764.20	793.29
Mean P [CPU] (W)	82.46	89.44	99.74
Total E CPUs+GPUs (kJ)	231	152	140
Total E entire system (kJ)	442	286	254
CO ₂ (gCO ₂ eq)	3.8	2.4	2.1
Accuracy (3 rounds)	0.556	0.557	0.563

Heterogeneous node usage correctly logged.
DVFS settings applied and verified.

SC2 - Large-scale

Table 3: Settings

Server: gros-104
Clients: 89 nodes (gros)
Freq. GHz: $f_{\max}$: 2.2, $f_{\min}$: 1
Framework: Flower
Strategy: FedAvg
Library: TensorFlow
Model: CustomCNN
Local Epochs: 1
Rounds: 5
Repetitions: 1

Table 4: Results

CPU Frequency (GHz)	$f_{\min}$ (1)	$f_{\max}$ (2.2)
Time monitored (s)	502.30	231.78
Time FL processed (s)	468.59	220.11
Mean power (Kwollect) (W)	69.50	76.14
Total energy (RAPL) (kJ)	919.8	565.4
Total energy (Kwollect) (kJ)	3254.4	1630.8
CO ₂ (gCO ₂ eq)	28.02	14.04
Accuracy	0.10	0.10

Total energy at $f_{\min}$ nearly $2 \times f_{\max}$ (idle power across 89 nodes dominates longer runs).
Framework stable at **90 nodes**.

SC3 - Cross framework

Table 5: Settings

Server: nova-14

Clients: nova-[15,16,17]

Freq. GHz: $f_{\min}=1.2$,
 $f_{\text{int}}=\{1.5,1.8\}$, $f_{\max}=2.1$

Framework: Flower*, FedScale⁺
Strategy: FedAvg

Library: PyTorch

Model: CustomCNN, ResNet18,
MobileNetV2

Local Epochs: 1* / 512⁺

Rounds: 10

Repetitions: 5

Energy Consumption to Reach Target Accuracy under Different Configurations

Training Model · FL Frameworks · CPU Frequencies

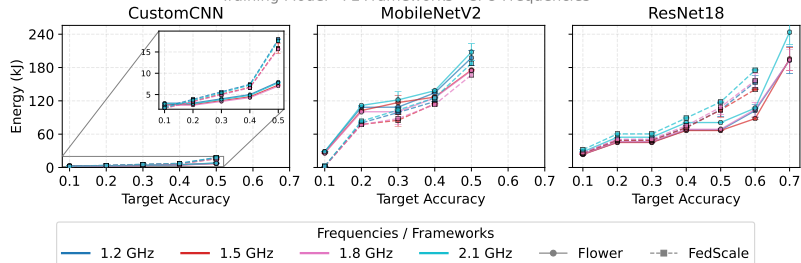


Figure 3: Energy vs. target accuracy across configs (SC3).

Energy remains relatively stable at low accuracy targets but increases sharply beyond a threshold → plan accuracy target carefully.

FedE-ator paper

- Title: **FedE-ator: A Reproducible Framework for Studying Energy Consumption in Federated Learning Systems**
- Accepted to the PECS workshop⁶ in the EUROPAR conference (2026).

⁶<https://pecs-workshop.github.io/2026/>

Research output summary

- The FedE-ator framework (version 1)⁷: manages heterogeneous HW (CPUs and GPUs) in one run + supports different FL frameworks.
- 1 paper accepted to a workshop + 1 paper submitted to a journal.

⁷<https://gitlab.irit.fr/sepia-pub/delight/eflwr>

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Planning for the coming year

- Upgrade FedE-ator with online DVFS-based energy efficiency for adaptive resource management (target: September 2026)
 - ▷ FL phase-aware adaptation
 - ▷ FL training-time-aware adaptation
 - ▷ CPU utilization-aware adaptation (system load balancing)
- Submit an article on the upgraded framework.
- Thesis preparation (starting in September 2026, expected defense in March 2027).

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Difficulties encountered

No major difficulties, work is progressing as planned.

Teaching

Tutorials for TP and TD sessions of the CSA class⁸ in the *Parallel* course (26h):

- **Monitoring**
- **MPI**

⁸class in English, for international M1 students

Training Progress

Category	Training activity	Date	Hours
Quality of life at work	<i>Débat théâtral "Entre les lignes"</i>	Apr. 2025	3
Dissemination of knowledge	<i>Journée Doctorants ASR</i>	Jun. 2024	8
	<i>MOOC: Reproducible Research II: Practices and Tools for Managing Computations and Data (Inria)</i>	Sep. 2024	10
Ethics and Scientific Integrity	<i>MOOC: Éthique de la recherche (Université de Lyon)</i>	Apr. 2026	6
	<i>MOOC: Intégrité scientifique dans les métiers de la recherche (Université de Bordeaux)</i>	Aug. 2025	6
	<i>MOOC: Recherche reproductible : principes méthodologiques pour une science transparente</i>	Jul. 2024	6
International / Languages	<i>Cours de Français Langue Étrangère (FLE) (Grade: 14/20)</i>	Jun. 2025	23
Interdisciplinarity / Scientific	<i>EcoICT Fall School 2024</i>	Oct. 2024	20
	<i>Summer School Green IT 2026</i>	Jun. 2026	34
	<i>What do we mean by Digital Sobriety?</i>	Oct. 2024	2
Career progression	<i>PhDOOC: PhD and Career Development</i>	Mar. 2025	12
Total			130

Merci pour votre attention!

Questions et discussion

Appendix: Research Questions

Research questions	Analysis objectives
Q1: What are the goals and proposals in the current studies?	Identify study objectives and the role of energy (constraint, optimization objective, or secondary effect).
Q2: What scope of energy consumption is taken into account?	Analyze energy measurement scope: computation, communication, end-to-end cost, and monitored resources (CPU, GPU, network, storage).
Q3: Which factors are considered to evaluate FL performance?	Identify evaluation metrics for performance and energy, and their impact on comparison across studies.
Q4: What key aspects should be considered for energy-aware and sustainable FL design?	Highlight energy-saving strategies, system heterogeneity, fairness, and environmental concerns.
Q5: What experimental methods are used to validate the proposals? How is energy consumption measured or estimated?	Analyze validation methods, energy measurement tools, and reproducibility support.
Q6: What are the current challenges and future research suggestions?	Summarize research gaps, challenges, and future directions.

FedE-ator: Key Features & Requirements

DVFS Control

- Adjusts CPU/GPU frequencies per instance
- Restores defaults upon completion

Reproducibility

- JSON-driven configuration
- Multiple instances x repetitions in one campaign
- Fixed role-to-node mapping across runs

► Features

FL Framework Flexibility

- **Intra-framework:** easy scale clients by JSON config
- **Inter-framework:** switch between Flower, FedScale^a

^a<https://github.com/SymbioticLab/FedScale>

Infrastructure requirements

- SSH + MPI
- Linux-based distributed systems